



Azure OpenAI Service

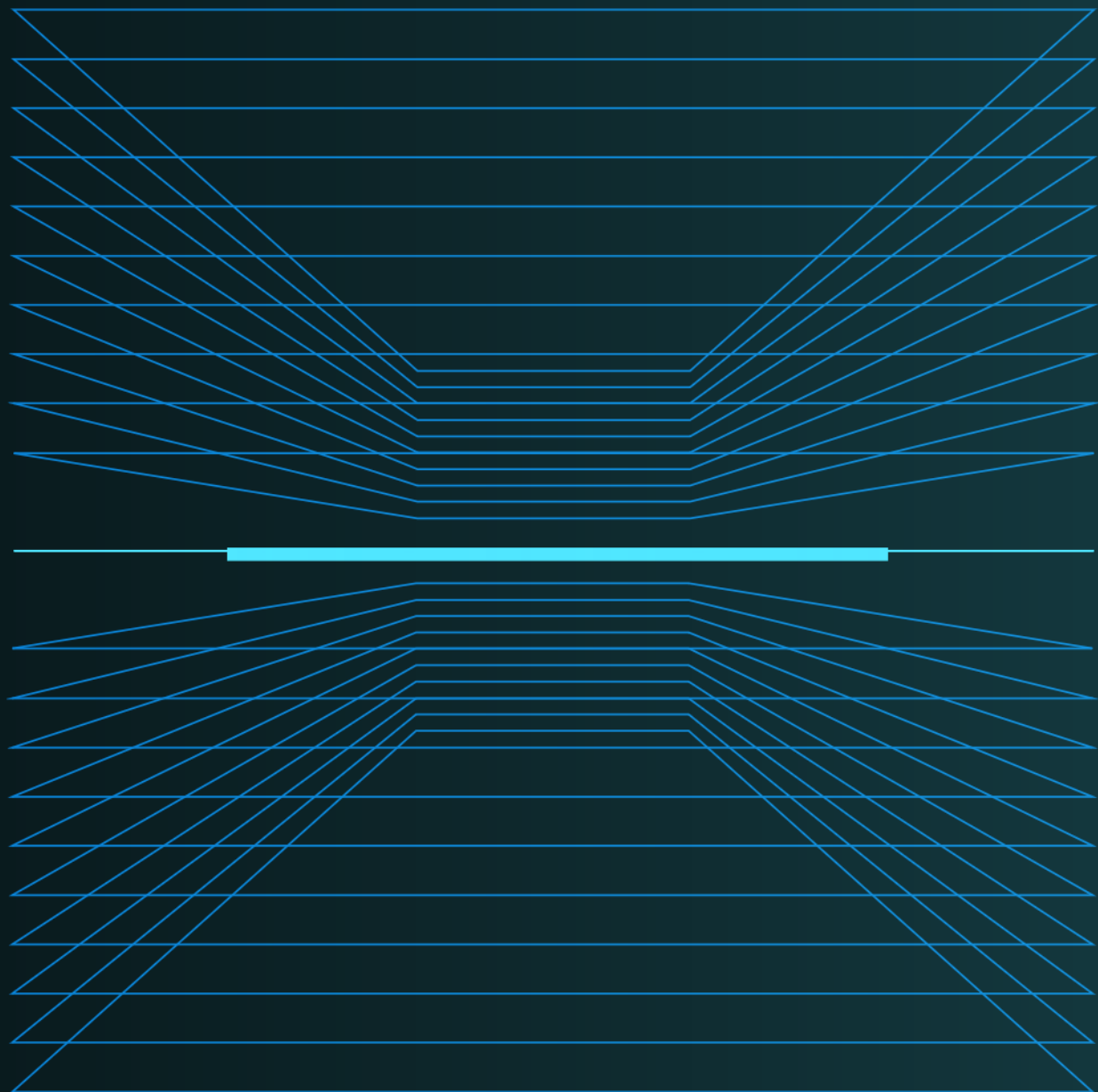
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September 18, 2024



Artificial Intelligence

Machine Learning

Deep Learning

Generative AI

1956

Artificial Intelligence

the field of computer science that seeks to create intelligent machines that can replicate or exceed human intelligence

1997

Machine Learning

subset of AI that enables machines to learn from existing data and improve upon that data to make decisions or predictions

2017

Deep Learning

a machine learning technique in which layers of neural networks are used to process data and make decisions

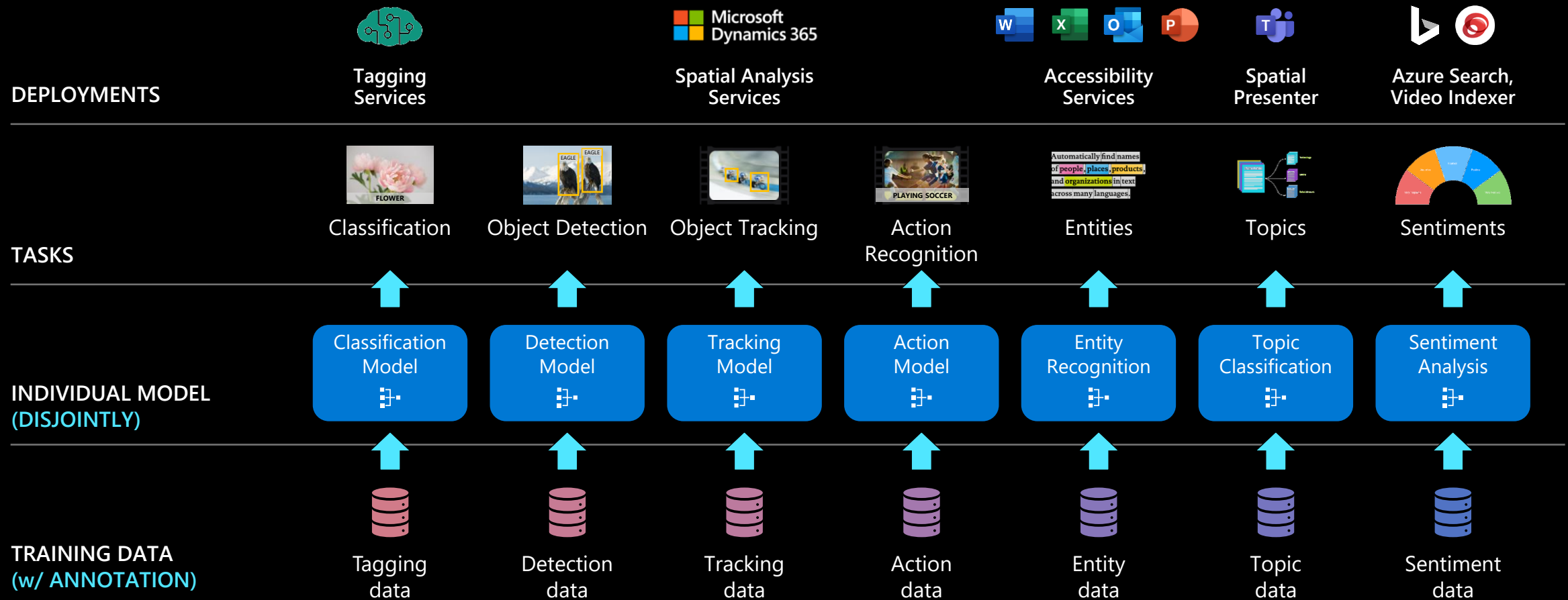
2021

Generative AI

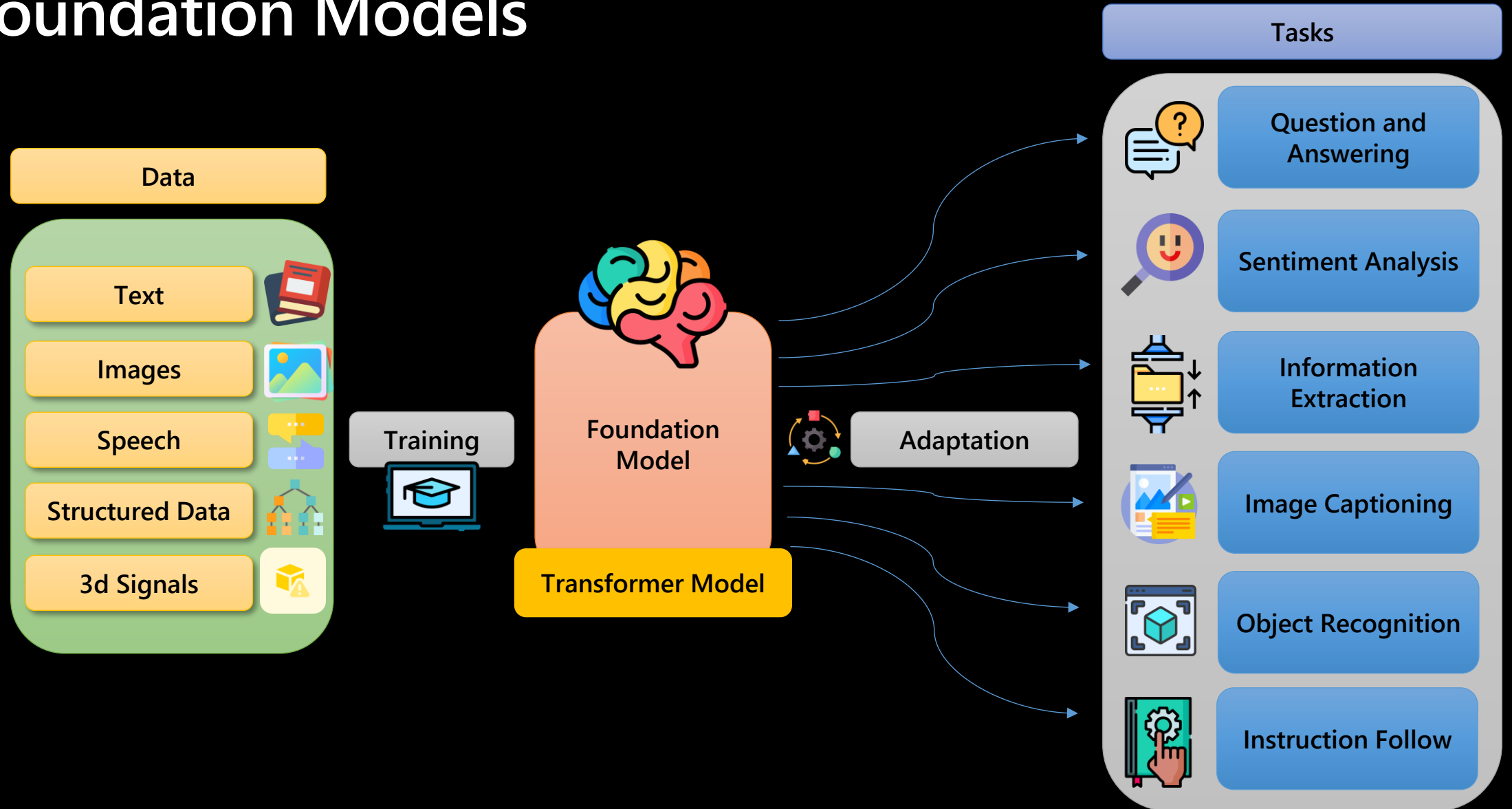
Create new written, visual, and auditory content given prompts or existing data.

Traditional model development

High cost and slow deployment—each service is trained disjointly



Foundation Models





OpenAI GPT-3

Why is the sky blue?

OpenAI GPT-3 answer:

The sky is blue because the Earth's atmosphere scatters sunlight in all directions and blue light is scattered more than the other colors because it travels as shorter, smaller waves

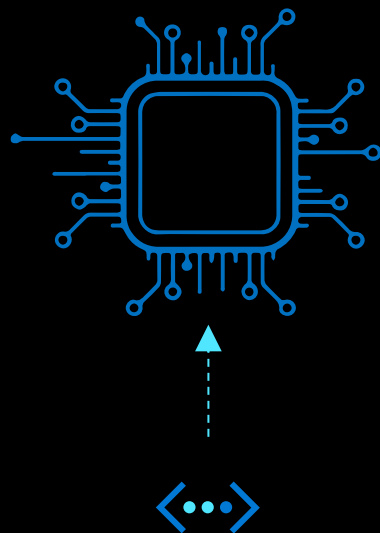


Don't fly solo.

Private code

**OpenAI Codex
Model**

GitHub



Public code and text
on the internet



**GitHub
Copilot Service**

Provide editor context

Provide suggestions

Improve suggestions

JS fetch_pic.js

push_to

```
1  const fetchNASAPictureOfTheDay = async () => {
2    return fetch('https://api.nasa.gov/image?api_key=DEMO_KEY&url=https://www.nasa.gov/images/content/108181main_108181main.jpg')
3      .then(response => response.json())
4      .then(json => {
5        method: 'GET',
6        headers: {
7          'Content-Type': 'application/json'
8        },
9      })
10     .then(response => response.json())
11     .then(json => {
12       return json;
13     });
14  }
```

Copilot

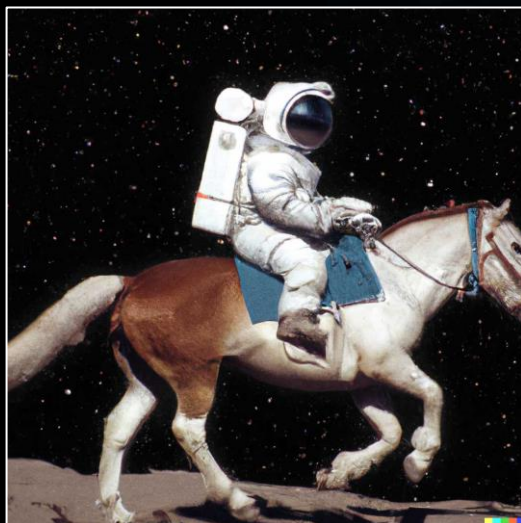




OpenAI

DALL·E 2

An astronaut riding a horse in a photorealistic style



Teddy bear working on new AI research on the moon in 1980



A bowl of soup that looks like a monster knitted out of wool





NUWA | Infinity

A cloudy morning
on the beach with
the tide coming in

Text to image

Image to video



Mountains
with clouds



Large Language Foundation models

Microsoft AI at Scale

Project Turing

17B
Turing-NLG | 2020

340M
BERT-Large | 2018



Our mission is to expand the boundaries of natural language understanding.

machine reading comprehension, question answering, transfer learning, reinforcement learning, computer vision, and even building interpretable models

Turing-NLG

Megatron NLG Turing

T-ULRv5

DeepSpeed

ZeRO

175B
GPT-3 | 2020


OpenAI

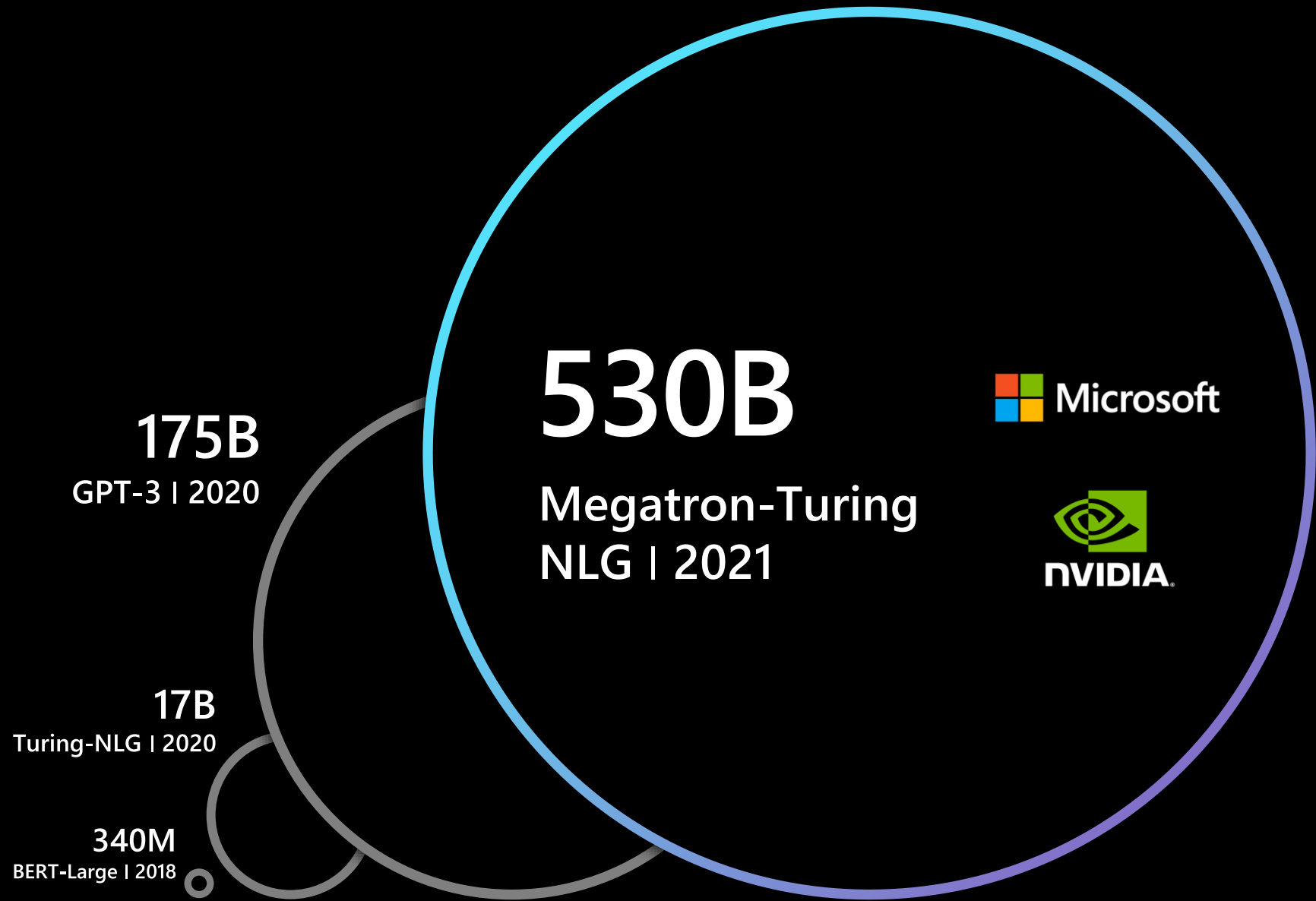
 Microsoft

17B
Turing-NLG | 2020

340M
BERT-Large | 2018

 Microsoft





Our partnership with OpenAI



Ensure that artificial
general intelligence (AGI)
benefits humanity.



Empower every person and
organization on the planet
to achieve more

GPT-3

Generate and Understand Text

Codex

Generate and Understand Code

DALL·E

Generate images from text prompts

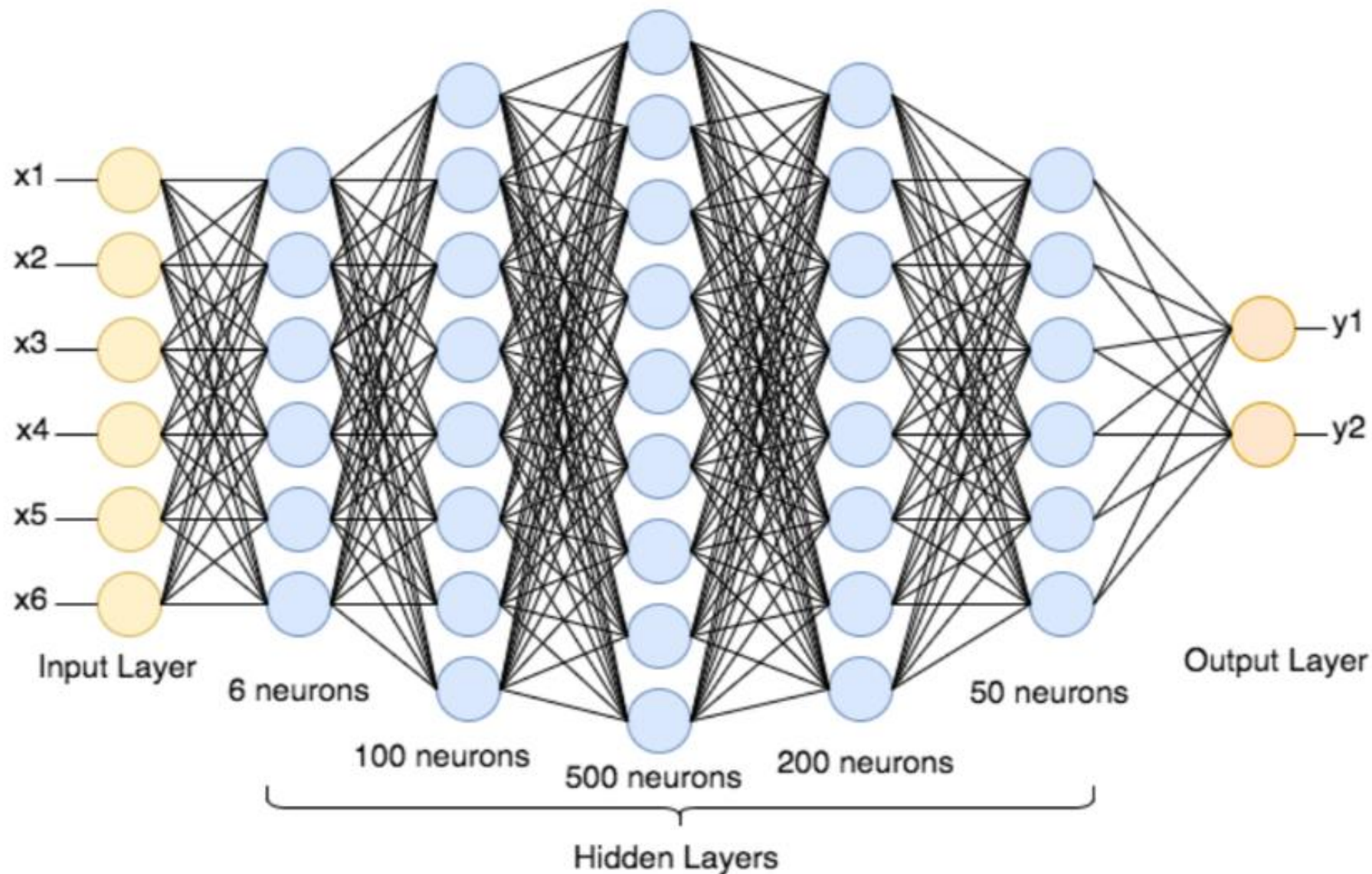
ChatGPT *(coming soon)*

Generate conversational responses

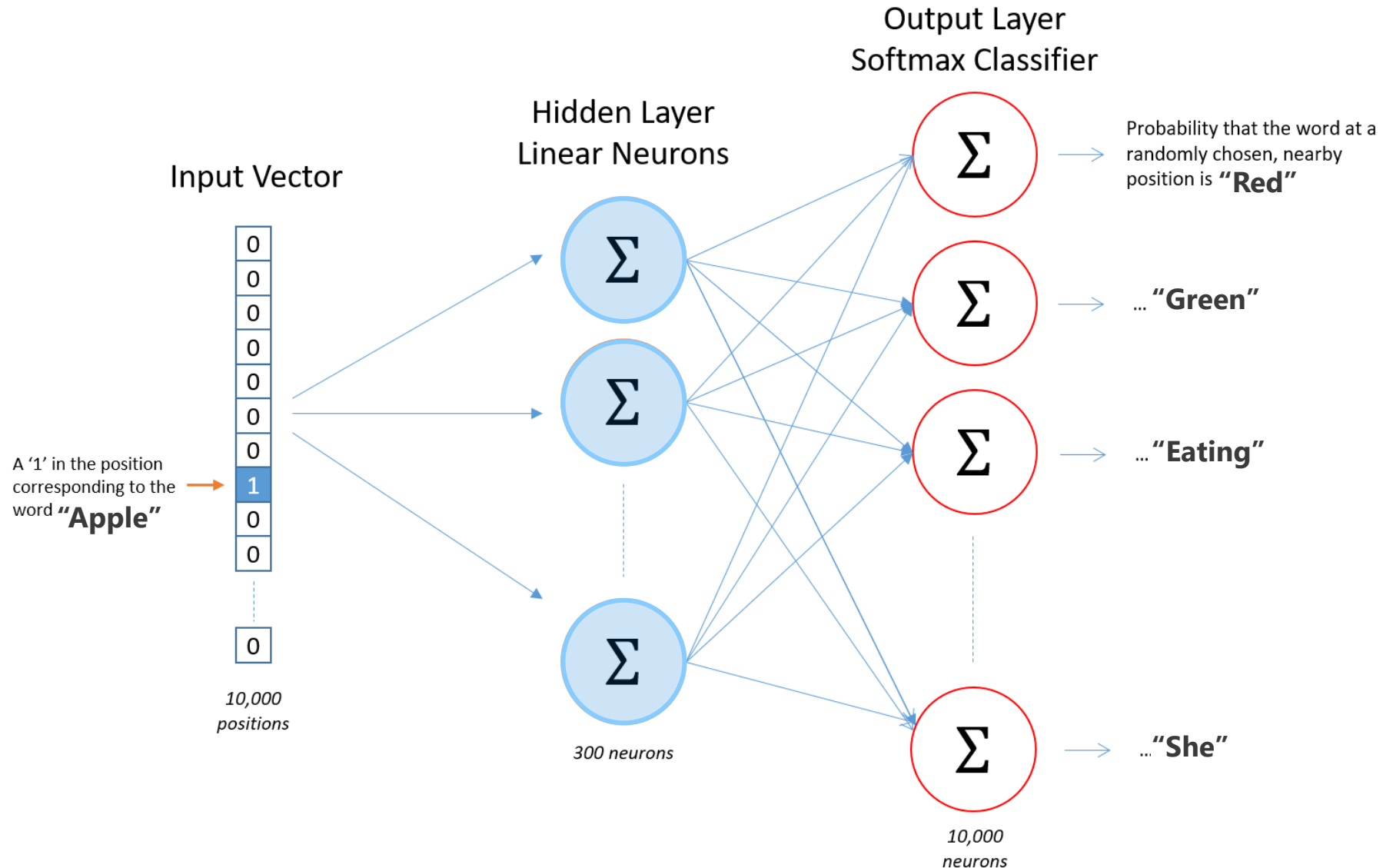
Learn to Describe Meaning to a Model

(Watching the blue dots)

"Apple"



A Close-up View of Learning



Describe the Word with Math (Vector Scores)

Input Vector

1 Corresponds to
the word
"Apple"

0	0	0	1	0
---	---	---	---	---

$\times$

Hidden-Layer
(blue dots)

$\longleftrightarrow E \longrightarrow$		
1	4	13
14	9	46
11	9	46
17	8	15
3	25	30
14	9	46
8	25	97
$\updownarrow V \updownarrow$		

$=$

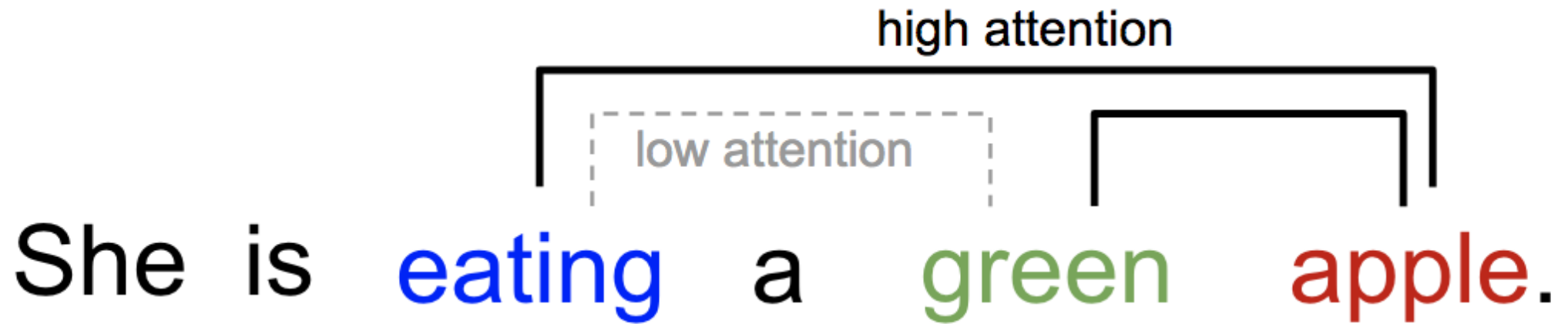
Word Embedding for
the word
"Apple"

17	8	15
----	---	----

Self-attention

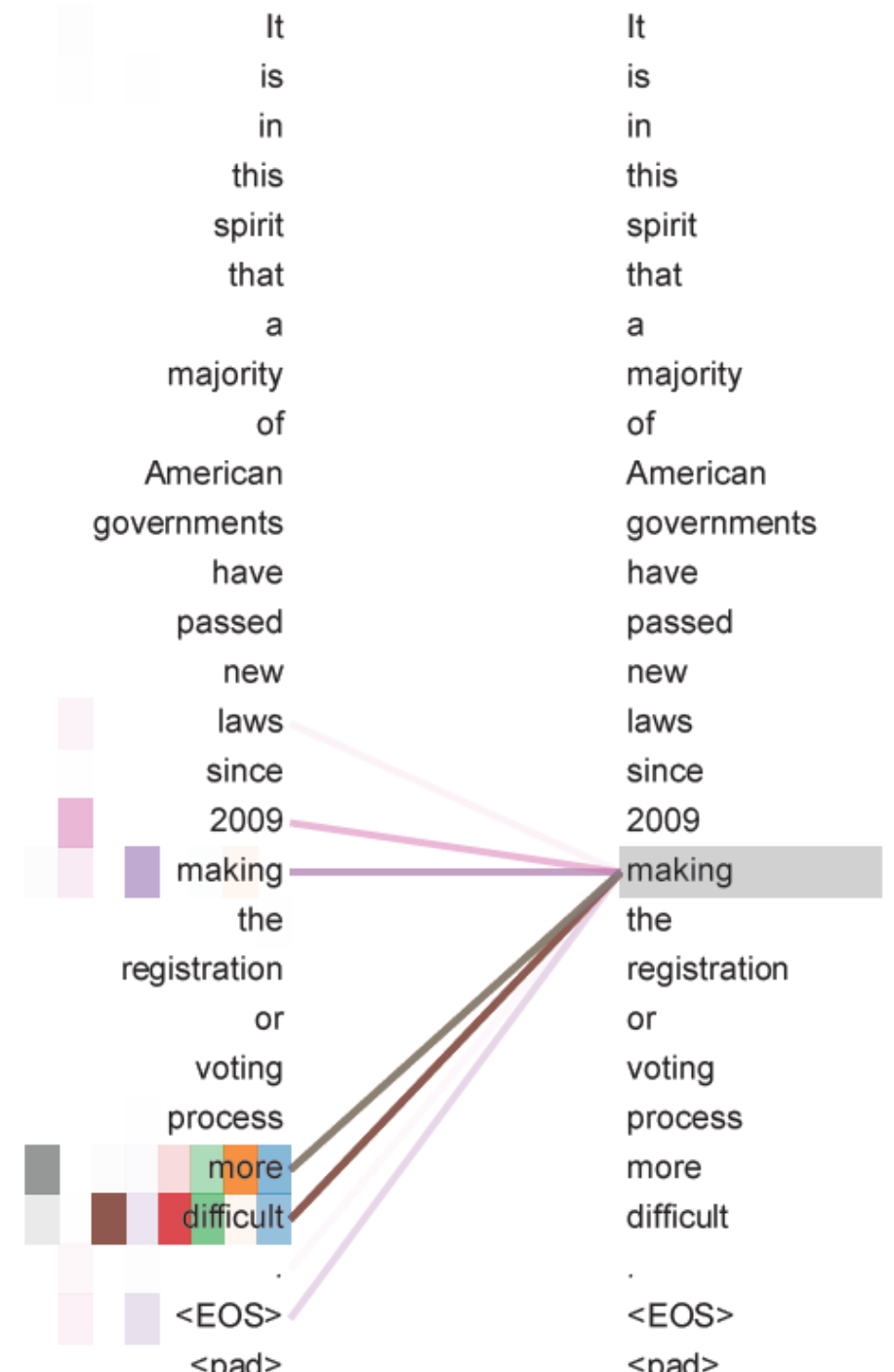
- This is the foundation of most of the splashy innovation we see in the popular press these days.
- We have seen how we can represent a character string as a numeric vector.
- Now imagine augmenting this vector with:
 - A positional marker for a sentence.
 - A score to indicate the connection between a word and surrounding words.

Attention: Connections Between Words

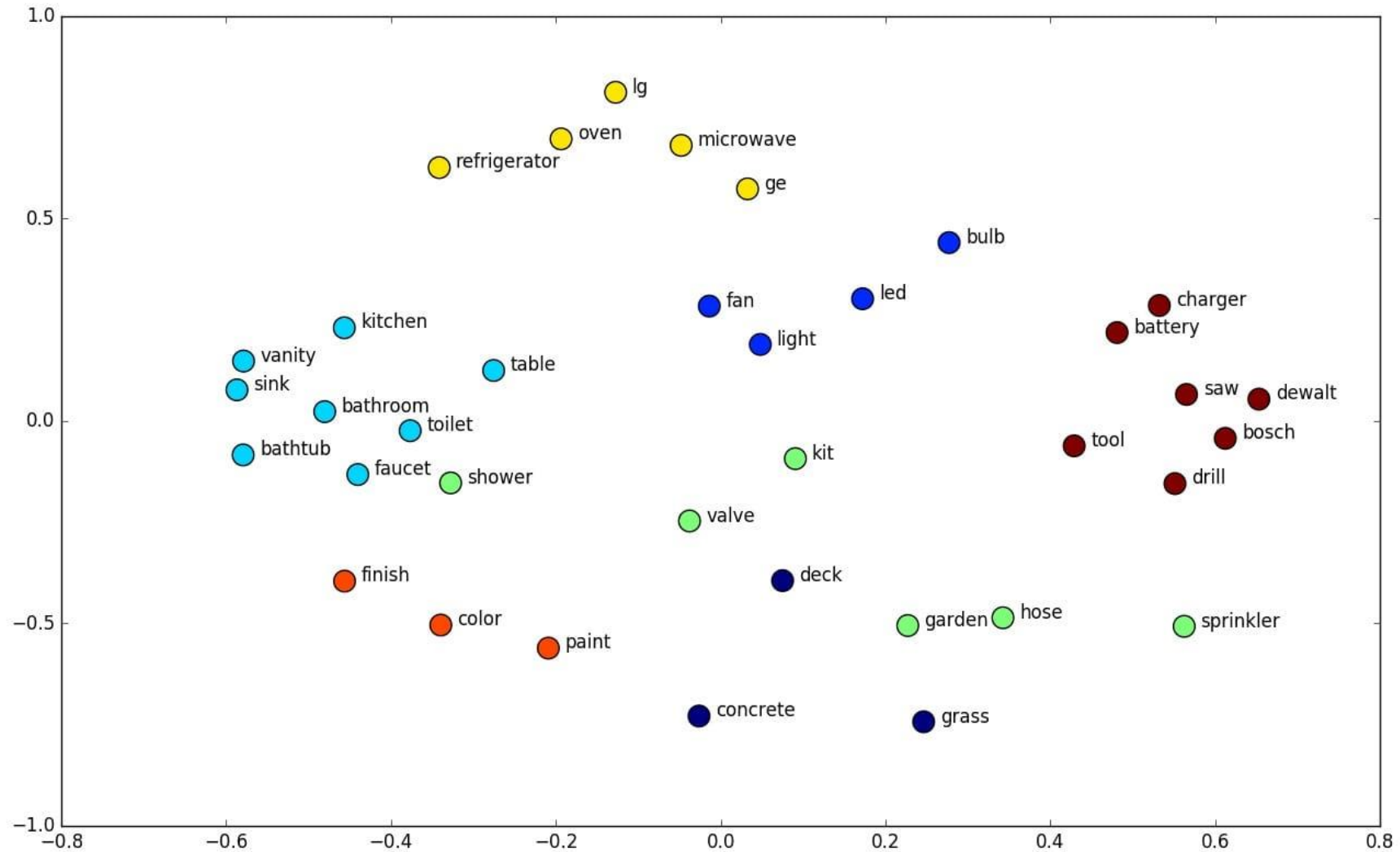


Attention visualizations

"It is in this spirit that a majority of American governments have passed new laws since 2009 making the registration or voting process more difficult."



Birds of a Feather, The Vectors Group Similar Words





Generative AI

GPT-3

Prompt:

Write a tagline for an ice cream shop.

Response:

We serve up smiles with every scoop!

Codex

Prompt:

```
Table customers, columns =  
[CustomerId, FirstName,  
LastName, Company, Address,  
City, State, Country,  
PostalCode]
```

Create a SQL query for all customers in Texas named Jane
query =

Response:

```
SELECT *  
FROM customers  
WHERE State = 'TX' AND  
FirstName = 'Jane'
```

DALL·E

Prompt: A white Siamese cat

Response:



ChatGPT *(coming soon)*

What is the fastest animal on land?

the cheetah (*Acinonyx jubatus*), which can reach speeds of up to 60 miles (97 kilometers) per hour.

What makes them so fast?

Cheetahs are built for speed and have several adaptations that make them the fastest land animal: lean body, long legs, flexible spine, large nostrils and claws that don't retract.



Azure OpenAI Service

GPT-3

Codex

DALL·E

ChatGPT *(coming soon)*



Deployed in your Azure subscription, secured by you, and tied to your datasets and applications



Large, pretrained AI models to unlock new scenarios



Custom AI models fine-tuned with your data and hyperparameters

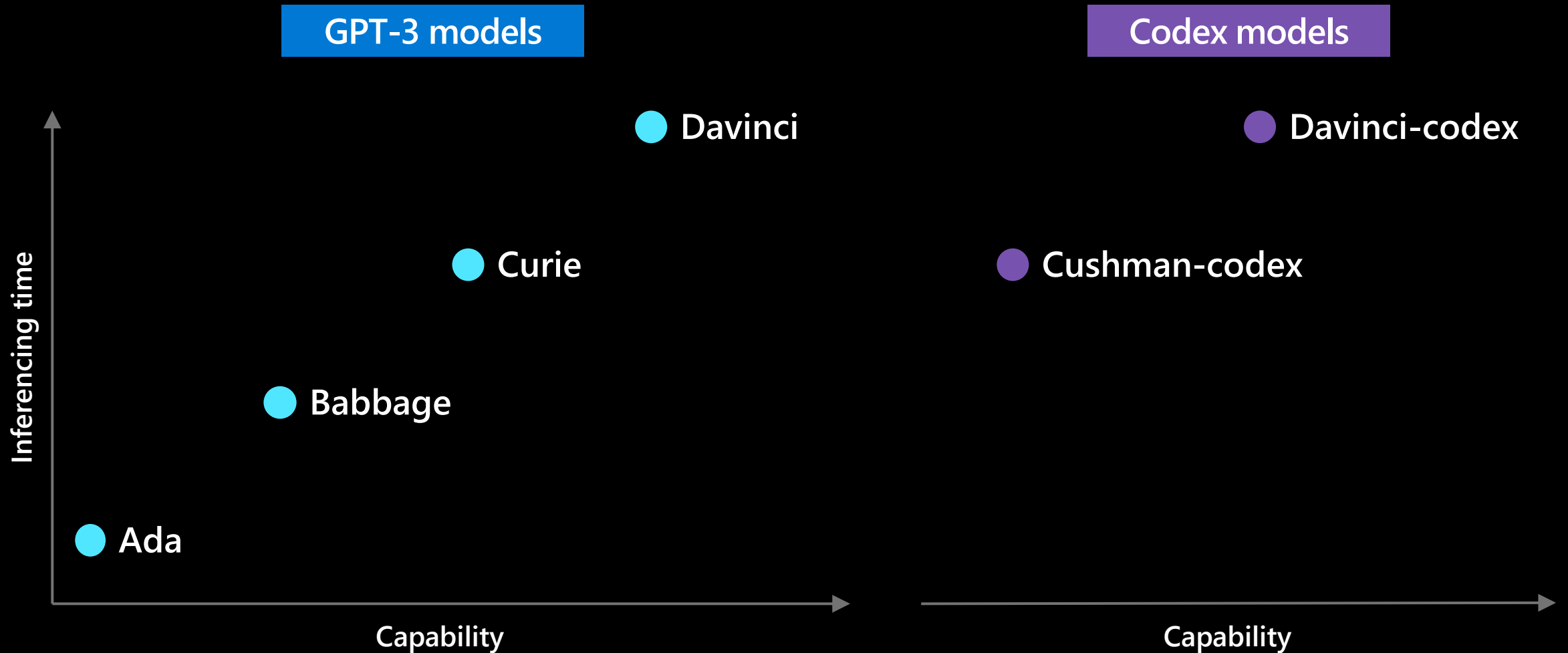


Built-in responsible AI to detect and mitigate harmful use



Enterprise-grade security with role-based access control (RBAC) and private networks

Azure OpenAI | Family of Models



| Azure OpenAI | GPT-3 Family of Models

Model	Max # Tokens per Request	Description, performance, cost	Use cases
Davinci	4,096 tokens	Most capable GPT-3 model. Can do any task the other models can do, often with <i>higher quality, longer output</i> and <i>better instruction-following</i> .	Complex intent, cause and effect, summarization for audience
Curie	2048 tokens	Very capable , but <i>faster</i> and <i>lower cost</i> than Davinci.	Language translation, complex classification, text sentiment, summarization
Babbage	2048 tokens	Capable of straightforward tasks, <i>very fast</i> , and <i>lower cost</i> .	Moderate classification, semantic search classification
Ada	2048 tokens	Capable of very simple tasks, usually the <i>fastest</i> model in the GPT-3 series, and <u>lowest cost</u> .	Parsing text, simple classification, address correction, keywords

| Azure OpenAI | Comparing the GPT-3.5 models

Model	Description
text-davinci-002	A GPT-3.5 model that was fine-tuned on natural language instructions and can perform a variety of tasks including summarization, question answering, classification, and more.
text-davinci-003	An improvement over the text-davinci-002 model. The model is similar to its predecessor but generally more capable across all tasks.
ChatGPT model (gpt-3.5) <i>(coming soon)</i>	A model fine-tuned from text-davinci-002 that was optimized for working with dialogue. ChatGPT is a great model to use for conversational tasks and also excels at creative tasks.

Visit the [Azure OpenAI Service pricing page](#) for pricing details

Azure OpenAI | Overview of GPT-3

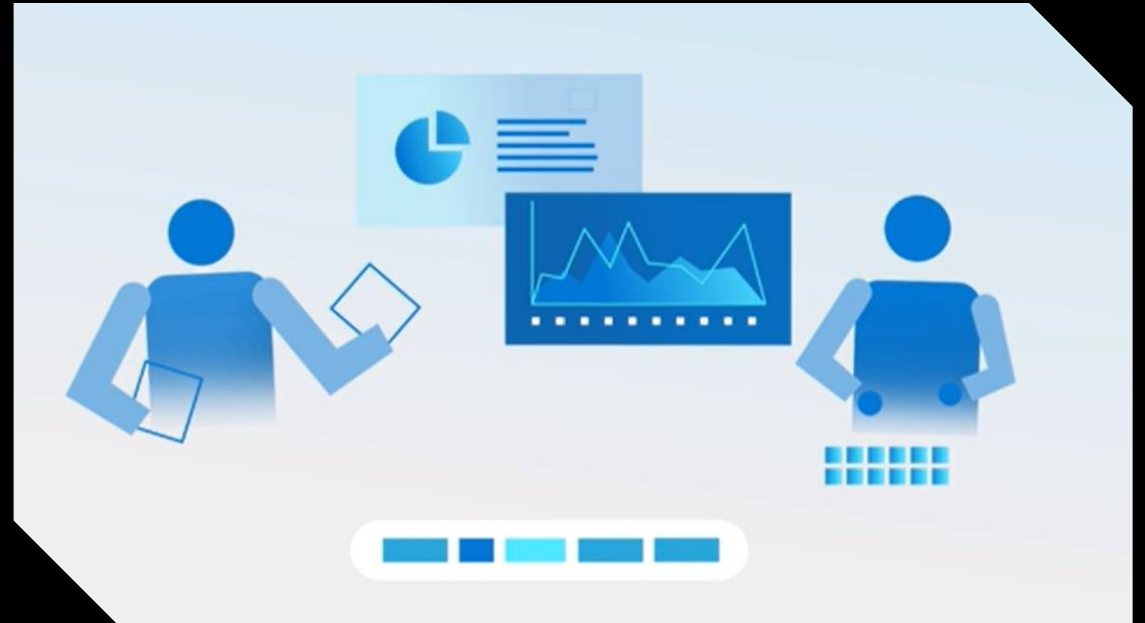
Generative pre-trained transformer 3 (GPT-3)

Autoregressive language model that uses deep learning to produce human-like text

Pre-trained on trillions of words

Predicts the most likely next word based on input text

General text-in/text-out interface



Azure OpenAI | GPT-3 Models

Powerful language
models accessible
to all skill levels



General purpose text-in/text-out
interface—flexibility



Simple UX—validate proof of concepts fast



Built in ML science intuition for everyone,
with deeper controls for ML practitioners

Azure OpenAI | GPT-3 Prompt Design

Extract the mailing address from this email:

Hi John Doe,

It was great to meet up at Build earlier this week. I thought the AI platform talk was great and I really enjoyed it.

I appreciate the offer for the book. If you are OK, you can mail it to me at home, or 123 Microsoft Way, Bellevue WA 92004.

Regards,

Chris Hoder

Prompt—Text input that provides some context to the engine on what is expecting.

Completion—Output that GPT-3 generates based on the prompt.

Azure OpenAI | GPT-3 Ideate, Experiment and Fine-Tune

Iterate on ideas with
a general-purpose
text-in/text-out interface

Prompt

Summarize game commentary
into highlights:

Shey Peddy is applying ball pressure at
the top against Sabrina Ionescu. At
7:48 remaining in the quarter; Peddy

What are the main highlights of the
game so far?

Sample response

The game has been close with Phoenix
leading New York 7-5. Shey Peddy has
been key for Phoenix.

Refine with examples
(‘few shot learning’) with
a simple UX

Prompt

Turn game commentary into highlights:

Commentary: What a pickup she has

Main highlights: New York has domina

###

Commentary:

1. Turner is so important defensively to
2. Griner pulled way out, Hartley with
3. At 1:54 remaining in the quarter, Pho

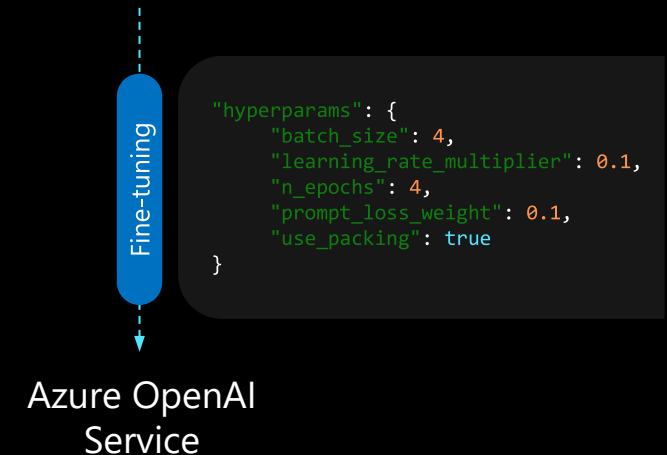
Sample response

Main highlights:

1. New York has had a strong run in the
2. Phoenix leading by 1 point, 24-23
3. New York Liberty's comeback has be

Optimize accuracy and
latency to validate proof
of concept fast

Prompt and completion examples



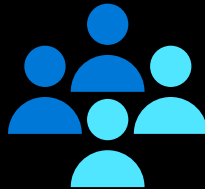
Codex Models



Makes APIs more accessible



Accelerates software development



Widens who can code

Codex Models

Derived from base models and trained on both NL and code (billions of Lines of Code)

Support multiple programming languages

- Python, C#, SQL, Java, JavaScript, TypeScript, Go, Perl, PHP, Ruby, Swift, Shell (bash)

Multiple tasks:

- Comment → Code
- Autocomplete function or next line (in context)
- Knowledge searching (API or Library call)
- Documenting code (comments)
- Refactoring

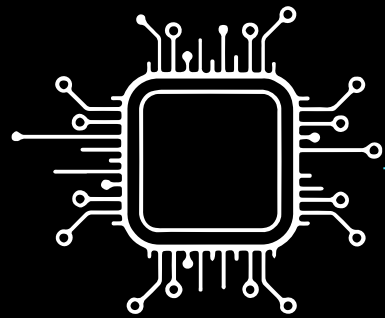
Use Cases

- Natural Language to Code
- Documenting code (comments)
- Refactoring
- Code to Natural Language
- Natural Language to SQL

Codex Models

```
19     screenshot = imageGrab.grab()
20     # Convert to text
21     text = image_to_string(screenshot)
22     # Parse text for email addresses
23     emails = re.findall(r'[\w\.-]+@[\w\.-]+', text)
24     return emails
25
26 def validate(addresses):
27     """
28
```

OpenAI Codex Model



Public code and text
on the internet

GitHub



GitHub
Copilot Service

Provide editor context

Provide suggestions

Improve suggestions

Private code

JS fetch_pic.js

push_to

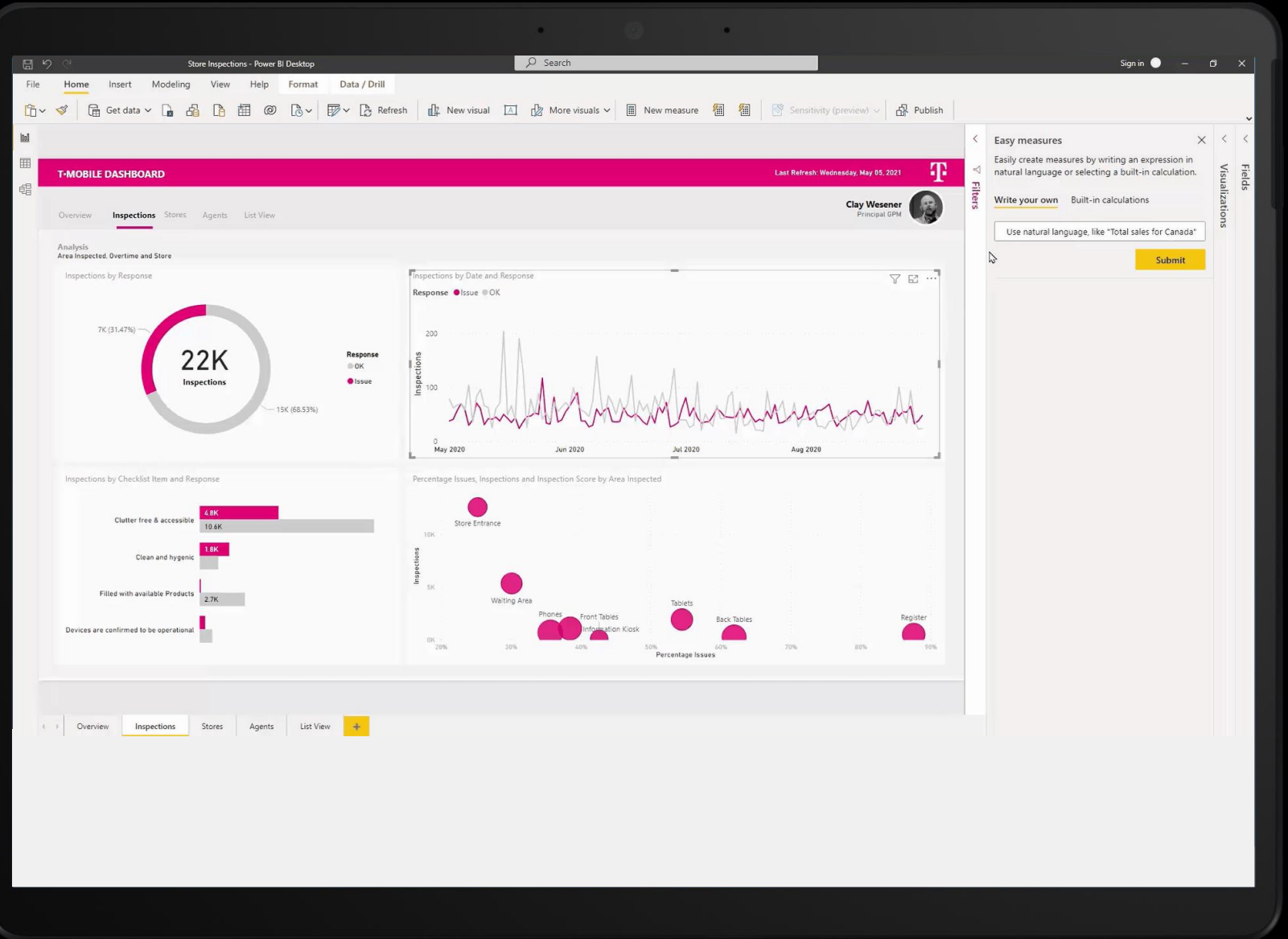
```
1  const fetchNASAPictureOfTh
2  return fetch('https://a
3    method: 'GET',
4    headers: {
5      'Content-Type': 'app
6    },
7  })
8  .then(response => resp
9  .then(json => {
10    return json;
11  });
12 }
```

Copilot



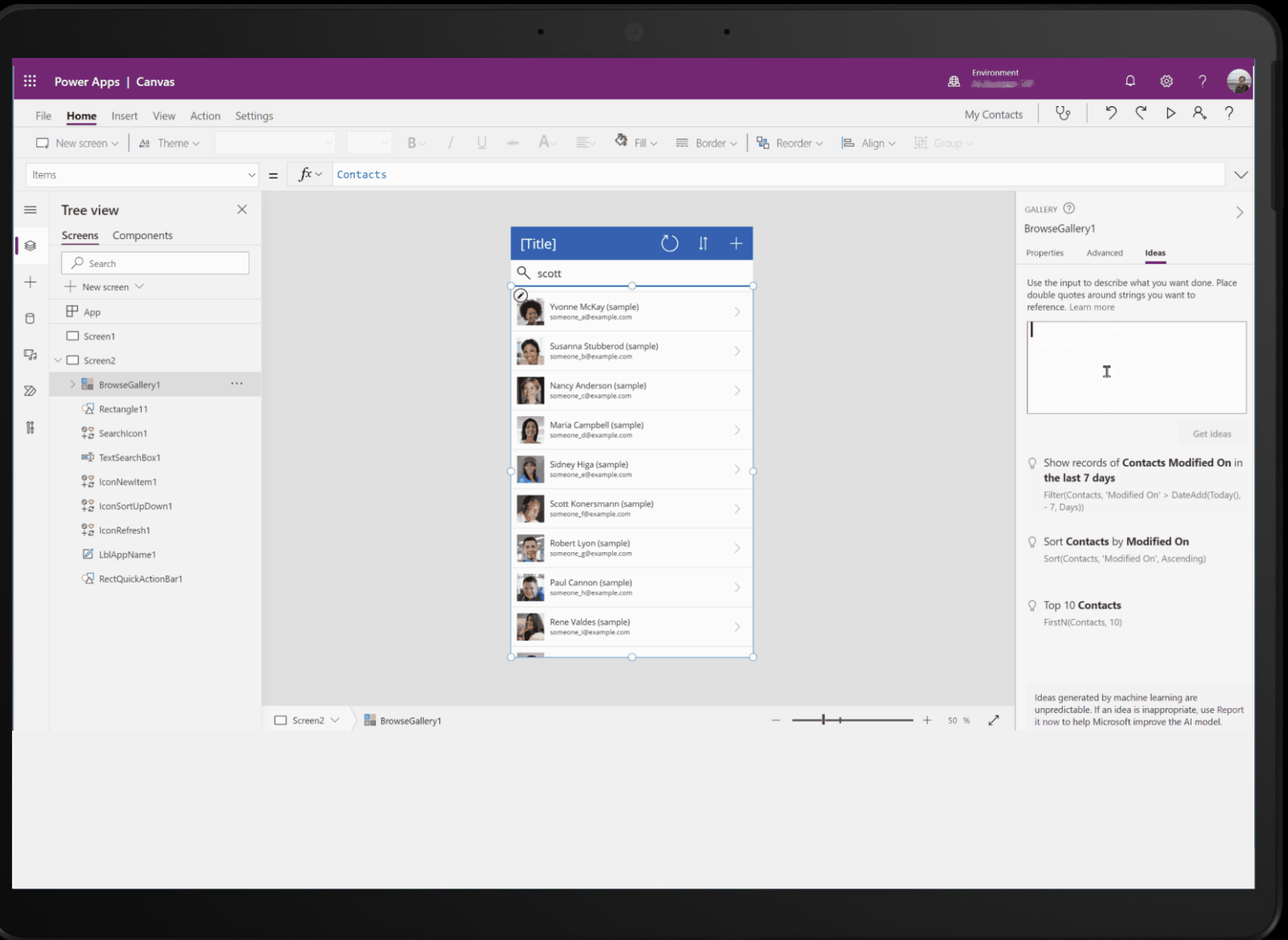
Power BI

Generating DAX expressions with natural language



Power Platform

Power Fx formulas



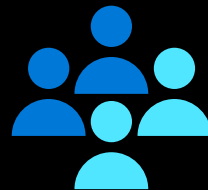
DALL·E 2



Generate an infinite number of images with simple text prompts



Accelerate designs or inspire creative decision



Build capability into enterprise applications through APIs and SDKs



Protection of species on our doorstep

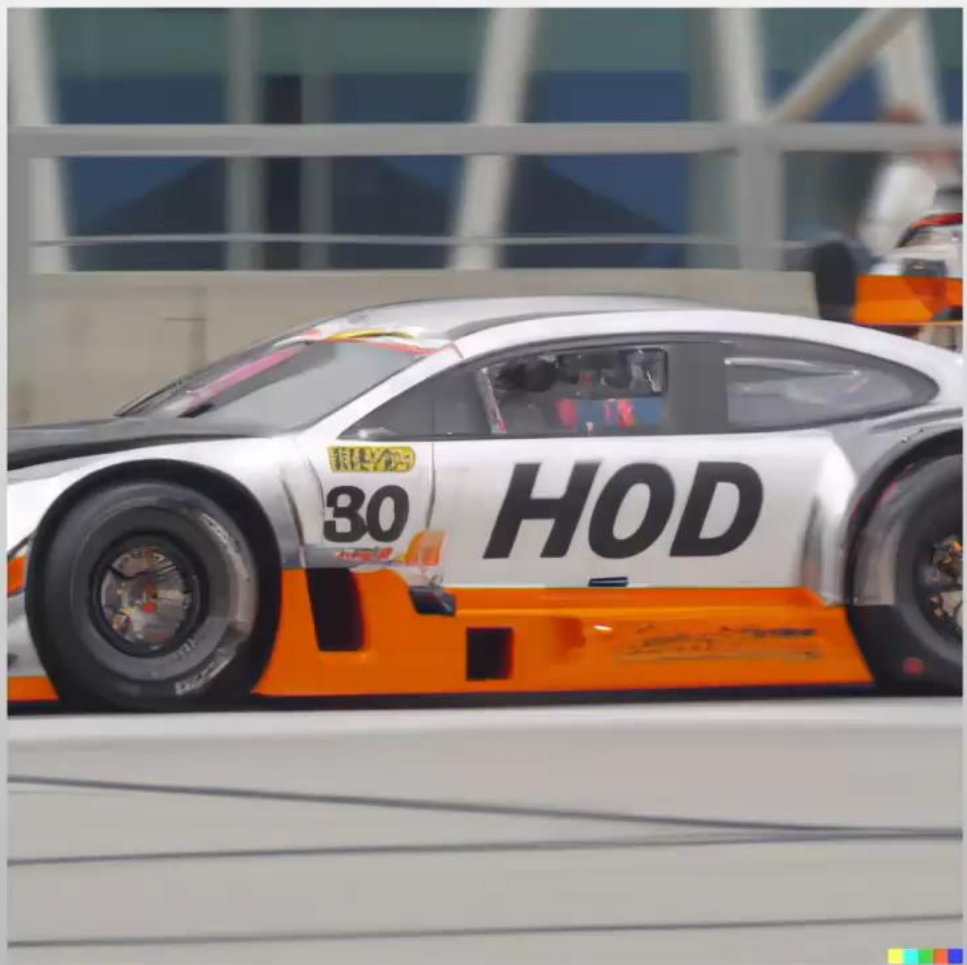
[GEOlino](#)

Episode 14 • 04.08.2021

25:48 min left

▶ Continue

We humans are causing a loss of species with our way of life and in this episode, we show you ways in which you can help. Whether by building nesting spaces or counting hamster burrows, get ready to roll up our sleeves and support the endangered species!



A DTM race car like a hot rod

Let's start with your content

< Generate an image using a description ⓘ

Ombre cake decorated with flowers and fall foliage



Tip
Explore different camera angles
"An aerial still of a seascape, Brazil"

Color Size



Continue editing Share

Azure OpenAI | When to use it



I need a general-purpose model that can handle multiple tasks.
e.g., translation+entity recognition+sentiment analysis



I need to generate human-like content, whilst preserving data privacy and security
e.g., abstractive summarization, content writing, paraphrasing, code



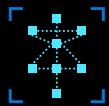
I need rapid prototyping and quick time to market for many use cases



I could use a model with little or no additional training



I want to explore solutions / use cases that have been described previously



Vision



Speech

Azure OpenAI Service



Language



Decision

Azure Cognitive Services



OpenAI

Developer experience & applicability

A broad set of text/code/image
generative use-cases:

- Flexible generative text, code and image API
- Popular examples scenarios—Copywriting, summarization, chatbots, code generation, image generation in design tools
- Experimentation with newest alpha models

Pricing

- Priced per 1K tokens. Training and inference
- Optimized for prototyping

Production- and enterprise-readiness

Limited enterprise-grade capabilities



Azure OpenAI Service

Everything stated for OpenAI, plus:

- Production-ready workloads requiring capacity
- Apps with global user-base requiring geo-diversity
- Apps requiring built-in content filtering and RAI checks
- Apps with compliance, data privacy and security needs

- Priced per 1K tokens. Training, inference & hosting
- Fine-tuned models optimized for production

GA—2023

Azure Active Dir, Data residency & compliance, data security, Azure SLA guarantee, global availability zones, Responsible AI



Cognitive Service for Language

Widely known language use-cases:

- Extraction, classification, chatbots, and summarization with formal support for multiple languages
- Translate text, and docs of any format—often used for web page localization

Projects requiring quick turn-around; minimal tuning

- Service priced per 1K text records. Predictable.
- Very competitively priced

Currently GA

Azure Active Dir, Data residency & compliance, data security, Azure SLA guarantee, global availability zones, Responsible AI

Azure OpenAI | Benefits

Increased efficiency and productivity by getting more done in lesser time through rapid prototyping and quicker time to market*

Faster time to realize value

Enhanced customer experience with a greater focus on customer-centric services and products, whilst utilizing feedback and trends better

Easy to use:
Even junior data scientists or business users can use Azure OpenAI Service Playground

Ensure data privacy & security, and implement it in a responsible manner using a Trusted Cloud Provider

Does not require a long tedious annotation process*

Less training data is required for many relevant use cases*

Ability to perform text analytics and generation tasks that up until now were reserved only to humans

* in most relevant use cases observed

Unlock new use cases



Content generation

Call Center Analytics: Automatically generate responses to customer inquiries

Government agency using Azure OpenAI Service to extract and summarize key information from their extensive library of rural development reports

Financial services using Azure OpenAI Service to summarize financial reporting for peer risk analysis and customer conversation summarization



Code generation

Aircraft company using to convert natural language to SQL for aircraft telemetry data

Consulting service using Azure OpenAI Service to convert natural language to query propriety data models



Semantic search

Financial services firm using Azure OpenAI Service to improve search capabilities and the conversational quality of a customer's Bot experience.

Insurance companies extract information from volumes of unstructured data to automate claim handling processes



Summarization

International insurance company using Azure OpenAI Service to provide summaries of call center customer support conversation logs

Global bank using Azure OpenAI Service to summarize financial reporting and analyst articles

Examples of multiple model use cases

End-to-end call center analytics

- Classification
- Sentiment
- Entity extraction
summarization
- Email generation

Customer 360

- Hyper-personalisation using
timely Summarization of
customer queries & trends
- Search
- Content generation

Business process automation

- Search through structured &
unstructured documentation
- Generate Code to query
data models
- Content Generation



Azure OpenAI Service

Use cases

Codex

- Natural Language to Code
- Natural Language to SQL
- Code to Natural Language
- Code documentation
- Refactoring

DALL·E 2

- Creative ideation
- Podcast and music playlist images
- Content syndication
- Marketing campaign personalization
- Hyper-personalization



Azure OpenAI Service

Use cases

Language

Reason over structured and unstructured data:
Classification, Sentiment, Entity Extraction, Search

- Refined text-based search with embeddings
- Product feedback sentiment
- Customer and employee feedback classification
- Claims and risk analyses
- Support emails and call transcripts
- Social media trends

Writing assistance

- Marketing copy/email taglines
- Long format text
- Paragraphs from bullets



Azure OpenAI Service

Use cases

Language

Summarization

- Call center call transcripts
- Subject Matter Expert Documents
 - Competitive analysis
 - Peer Analysis
 - Technical reports
- Product and service feedback
- Social media trends

Conversational AI

- Smart assists for call centers
- Tech support chat bots
- Virtual assistants
- Artificial humans



Azure OpenAI Service

Use cases

Language

Use Cases that use multiple model capabilities

- Contact Centers
 - Classification—route mails to appropriate team
 - Sentiment—prioritize angry customers
 - Entity extraction and search—analyze liability and risk
 - Mail and call transcript summarization
 - Customer response email generation
- Rapid response marketing campaigns: classification, sentiment, summarization, content generation, image generation



Additional considerations

This is only a REST API. It needs to be connected to applications and/or business processes to be useful.

These models use input from instructions and examples in the prompt to identify the task.

The model completes the task by predicting the most probable next text.

In the next section(s), we will cover three main approaches for in-context learning.

There are some additional considerations when choosing a use case. [Learn more](#)

Naming Convention

Azure OpenAI's model names typically correspond to the following standard naming convention:

{family} - {capability} [-{input-type}] - {identifier}

Element	Description
{family}	The model family of the model. For example, <u>GPT-3 models</u> uses text, while <u>Codex models</u> use code.
{capability}	The relative capability of the model. For example, GPT-3 models include ada, babbage, curie, and davinci.
{input-type}	(<u>Embeddings models</u> only) The input type of the embedding supported by the model. For example, text search embedding models support doc and query.
{identifier}	The version identifier of the model.



This model is a GPT-3 text model, the most powerful (davinci), and of the latest version (003)

**<https://learn.microsoft.com/en-us/azure/cognitive-services/openai/concepts/models>*

Tokens

You can think of tokens as pieces of words used for natural language processing. For English text, 1 token is approximately 4 characters or 0.75 words.

As a point of reference, the collected works of Shakespeare are about 900,000 words or 1.2M tokens.

Understanding tokens and probabilities

Tokens:

I have an orange cat named Butterscotch.

I have an orange cat named Butterscotch.

Horses are my favorite

animal	49.65%
animals	42.58%
\n	3.49%
!	0.91%

Probabilities:

IF TEMPERATURE IS 0

Horses are my favorite animal

Horses are my favorite animal

Horses are my favorite animal

Horses are my favorite animal

IF TEMPERATURE IS 1

Horses are my favorite animal

Horses are my favorite animals

Horses are my favorite !

Horses are my favorite animal

Prompt Instruction

Suggest three names for an animal that is a superhero.

Animal: Cat
Names: Captain Sharpclaw, Agent Fluffball, The Incredible Feline
Animal: Dog
Names: Ruff the Protector, Wonder Canine, Sir Barks-a-Lot
Animal: Horse
Names:

Completion Temperature 0 (always the same)

Mighty Equine, The Great Galloper, Thunderhoof

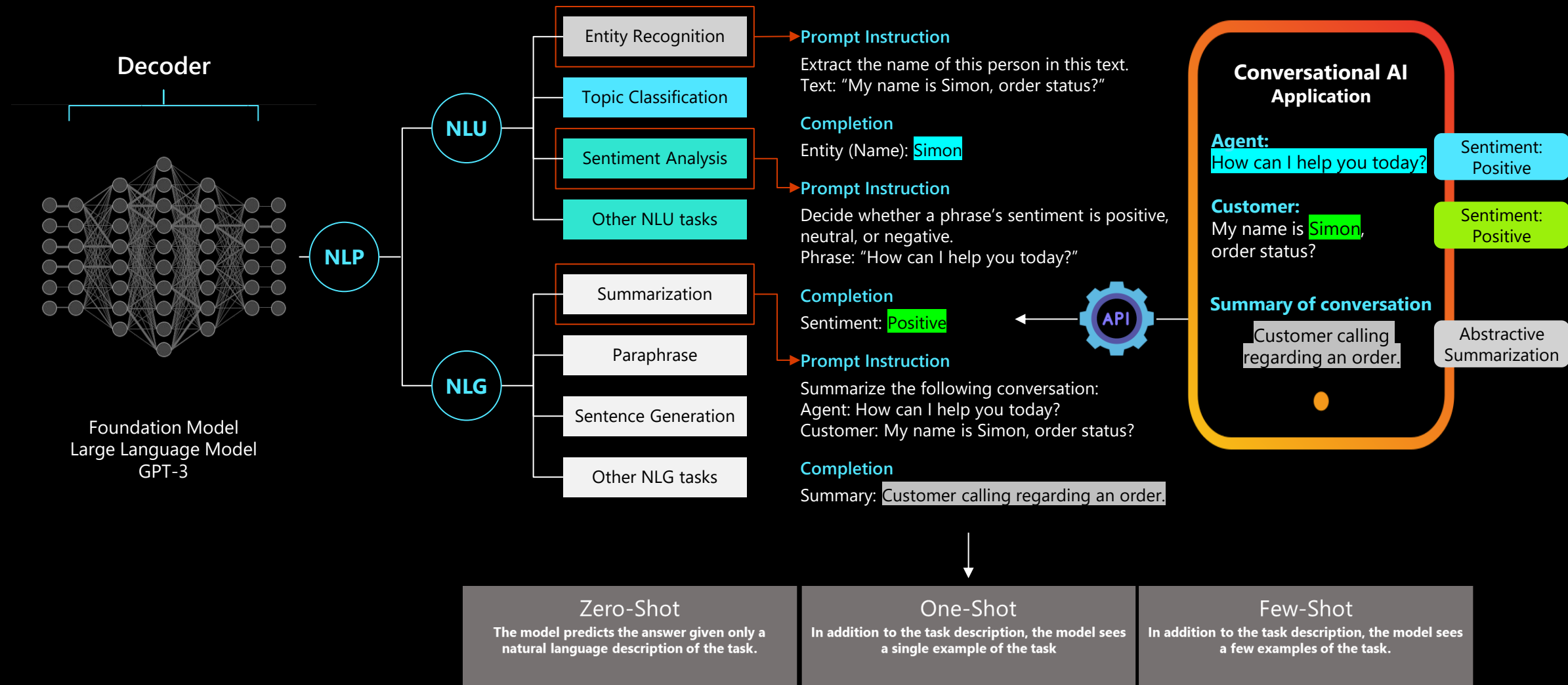
Completion Temperature 1 (often different)

Blaze the Miracle Mare, Pegasus the Winged Warrior, Secretariat the Superhorse

Completion Temperature 1 (often different)

Blaze of Glory, Sterling Silver, Thunderbolt

Model use out of the box—prompting



Text classification

Named Entity Recognition (Zero-Shot)

Prompt:

Extract the name and mailing address from this email:

Dear Kelly,

It was great to talk to you at the seminar. I thought Jane's talk was quite good.

Thank you for the book. Here's my address 2111 Ash Lane, Crestview CA 92002

Best,

Maya

Completion:

Name: Maya

Mailing Address: 2111 Ash Lane, Crestview CA 92002

Text classification

Custom Named Entities (Few-Shot)

Prompt:

Extract job titles from the following sentences.

Sentence: John Doe has been working for Microsoft for 20 years as a Linux Engineer.

Job title: Linux Engineer

###

Sentence: John Doe has been working for Microsoft for 20 years and he loved it.

Job title: none

###

Sentence: Marc Simoncini | Director | Meetic

Job title: Director

###

Sentence: Damien is the CTO of Platform.sh, he was previously the CTO of Commerce Guys, a leading ecommerce provider.

Completion:

Job title: CTO

Text classification

Multi-Label Text Classification (Few-Shot)

Prompt:

The following is a list of movies and the categories they fall into:

Topgun: As students at the United States Navy's elite fighter weapons school compete to be best in the class, one daring young pilot learns a few things from a civilian instructor that are not taught in the classroom.

Category: Drama, Action

Hustle: A basketball scout discovers a phenomenal street ball player while in Spain and sees the prospect as his opportunity to get back into the NBA.

Category: Comedy, Drama, Sport

The Northman: From visionary director Robert Eggers comes The Northman, an action-filled epic that follows a young Viking prince on his quest to avenge his father's murder.

Category:

Completion :

Action, Adventure, Drama

Text classification

Paraphrasing (Few-Shot)

Prompt:

Article: Searching a specific search tree for a binary key can be programmed recursively or iteratively.

Paraphrase: Searching a specific search tree according to a binary key can be recursively or iteratively programmed.

Article: It was first released as a knapweed biocontrol in the 1980s in Oregon , and it is currently established in the Pacific Northwest.

Paraphrase: It was first released as Knopweed Biocontrol in Oregon in the 1980s , and is currently established in the Pacific Northwest.

Article: 4-OHT binds to ER , the ER / tamoxifen complex recruits other proteins known as co-repressors and then binds to DNA to modulate gene expression.

Paraphrase: The ER / Tamoxifen complex binds other proteins known as co-repressors and then binds to DNA to modulate gene expression

Article: Google was founded in 1998 by Larry Page and Sergey Brin while they were Ph.D. students at Stanford University in California. Together they own about 14 percent of its shares and control 56 percent of the stockholder voting power through supervoting stock. They incorporated Google as a privately held company on September 4, 1998

Completion:

Paraphrase: Google was incorporated as a privately held company on September 4, 1998 by Larry Page and Sergey Brin, while they were Ph.D. students at Stanford University in California. They own about 14 percent of its shares and control 56 percent of the stockholder voting power through supervoting stock.

Reason over unstructured text

Question and answering

Prompt:

On Monday, Microsoft launched an OpenAI service as part of its Azure cloud platform, offering businesses and start-ups the ability to incorporate models like ChatGPT into their own systems. The company has already been building AI tools into many of its consumer products, such as a DALL-E 2 feature in its Bing search engine that can create images based on a text prompt, and the Information reported recently that it's working to bring more of them to Microsoft Office as well.

What will Microsoft incorporate in its own systems?

Completion:

Microsoft plans to incorporate AI tools into its own systems, such as a DALL-E 2 feature in its Bing search engine that can create images based on a text prompt, and the Information reported recently that it's working to bring more of them to Microsoft Office as well

Text classification

Keyword Extraction (Few-Shot)

Prompt:

Information Retrieval (IR) is the process of obtaining resources relevant to the information need. For instance, a search query on a web search engine can be an information need. The search engine can return web pages that represent relevant resources.

Keywords: searching, missing, desert

###

I believe that using a document about a topic that the readers know quite a bit about helps you understand if the resulting keyphrases are of quality.

Keywords: document, understand, keyphrases

###

Since transformer models have a token limit, you might run into some errors when inputting large documents. In that case, you could consider splitting up your document into paragraphs and mean pooling (taking the average of) the resulting vectors.

Completion:

Keywords: transformer, models, token

Summarization

Extractive vs Abstractive

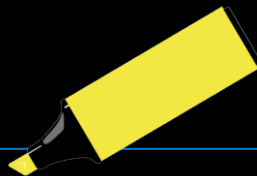
Extractive

Select parts of the original text to form a summary

- Easier
- Restrictive (no paraphrasing)

Peter and Elizabeth took a taxi to attend the night party in the city. While in the party, Elizabeth collapsed and was rushed to the hospital.

Peter and Elizabeth attend party city. Elizabeth rushed hospital



Extractive (Semantic)

Generates new text using natural language generation techniques

- More difficult
- More flexible (more human)

Peter and Elizabeth took a taxi to attend the night party in the city. While in the party, Elizabeth collapsed and was rushed to the hospital.

Elizabeth was hospitalized after attending a party with Peter.



Abstractive summarization

Contact Center Summarization

Prompt:

Summarize this for a call center agent:

Agent: Thank you for calling ADM. Who am I am speaking to?

Customer: Hello, my name is Peter Smith. I own a small business and have some questions regarding payroll processing.

Agent: Good morning, Peter, before we get started may I ask you a few questions so that I better answer your questions today?

Customer: Thank you that is quite helpful. Are there specific regulations that I need to follow?

Agent: Certain aspects of payroll processing are regulated by the Internal Revenue Service (IRS) and the Department of Labor (DOL)

Completion:

The agent is speaking to a customer named Peter Smith. Peter is asking for tips on manual payroll processing. The agent also informs Peter that certain aspects of payroll processing are regulated by the Internal Revenue Service (IRS) and the Department of Labor (DOL).

Prompt Insert

Only DaVinci-002+

Before

These are agenda topics for a customer presentation.

1. Introduction to GPT3

[insert]

10. Conclusion

After

These are agenda topics for a customer presentation.

1. Introduction to GPT3

2. Overview of GPT3 Features

3. Benefits of GPT3

4. GPT3 Use Cases

5. GPT3 Pricing

6. GPT3 Security

7. GPT3 Support

8. GPT3 Integration

9. Q&A Session

10. Conclusion

Intent classification (Few-Shot)

listen to WestBam album allergic on google music: PlayMusic

give me a list of movie times for films in the area: SearchScreeningEvent

show me the picture creatures of light and darkness: SearchCreativeWork

I would like to go to the popular bistro in oh: BookRestaurant

what is the weather like in the city of Frewen in the country of Venezuela: GetWeather

I want to book a flight to Delhi: bookFlight

Chatbot with personality (Few-Shot)

This is a discussion between a [human] and a [robot].
The [robot] is very nice and empathetic.

[human]: Hello nice to meet you.

[robot]: Nice to meet you too.

###

[human]: How is it going today?

[robot]: Not so bad, thank you! How about you?

###

[human]: I am ok, but I am a bit sad...

[robot]: Oh? Why that?

###

[human]: I broke up with my girlfriend...

[robot]: Oh? How did that happen?

Analyze customer feedback (Few-Shot)

Examples of customer reviews:

"The boyshorts work much better for me for sleep."

"I wish the boyshort came in a high waist option."

"You definitely can still use a pad but it doesn't improve the leak proof ness of it unless it's a high coverage style like the boyshort."

Feedback: The customer seems to like the boyshort style better than the bikini style for sleeping. They also wish that the high waist option was available for the boyshort style. The customer also believes that the leak proofness of the boyshort can be improved if it is a high coverage style.

Topic classification for Bot routing (Few-Shot)

Message: When the spaceship landed on Mars, the whole humanity was excited

Topic: space

###

Message: I love playing tennis and golf. I'm practicing twice a week.

Topic: sport

###

Message: Managing a team of sales people is a tough but rewarding job.

Topic: business

###

Message: I am trying to cook chicken with tomatoes

Topic: Cooking

Summarize conversational transcripts (Few-Shot)

Summarize this conversation transcript:

Agent: Thank you for calling ADM. Who am I am speaking to? **Customer:** Hello, my name is Peter Smith. I own a small business and have some questions regarding payroll processing. **Agent:** Good morning, Peter, before we get started may I ask you a few questions so that I better answer your questions today? **Customer:** Yes, of course. **Agent:** In case we get disconnected, can you please share your phone number and email address so that we may contact and share additional information? **Customer:** Yes, my phone number is (514) 777-5232 and my email address is petersmith@gmail.com **Agent:** May I ask where is your business located so I can answer your questions based on your location **Customer:** Yes, my business is located in Miami, Florida. **Agent:** Thank you, what can I help you with today? **Customer:** I am fairly new to payroll processing and do everything manually today. Do you have any advice or tips on manual payroll processing? **Agent:** If you're a small business with only a few employees and choose to process payroll manually, you will need to keep precise records of hours worked, wages paid and worker classifications, among other details. You must also ensure your calculations are correct and remember to file all the necessary taxes and paperwork with government authorities on time. **Customer:** Thank you that is quite helpful. Are there specific regulations that I need to follow? **Agent:** Certain aspects of payroll processing are regulated by the Internal Revenue Service (IRS) and the Department of Labor (DOL). Some of the laws you must comply with include: Fair Labor Standards Act (FLSA), Federal Insurance Contributions Act (FICA), Federal Unemployment Tax Act (FUTA) **Customer:** This sounds very complicated. I don't really feel like managing all of this. **Agent:** If you would like I can transfer you to one of our specialized agents who can talk to you about payroll service providers that would help you manage all of that. **Customer:** That would be great, thank you very much for al your help.

The agent is speaking to a customer named Peter Smith. Peter is asking for tips on manual payroll processing. The agent advises Peter to keep precise records and to file all necessary taxes and paperwork on time. The agent also informs Peter that certain aspects of payroll processing are regulated by the Internal Revenue Service (IRS) and the Department of Labor (DOL). Finally, the agent offers to transfer Peter to a specialized agent who can help him find a payroll service provider.

Product description and ad generation (Few-Shot)

Generate a product description out of keywords.

Keywords: shoes, women, \$59

Sentence: Beautiful shoes for women at the price of \$59.

###

Keywords: trousers, men, \$69

Sentence: Modern trousers for men, for \$69 only.

###

Keywords: gloves, winter, \$19

Sentence: Amazingly hot gloves for cold winters, at \$19.

###

Keywords: t-shirt, men, \$39

Sentence:

Sentence: Extraordinary t-shirt for men, for \$39 only.

Product recommendation (Few-Shot)

Generate product recommendations:

Product: table, chair, plate

Recommendation: utensils

###

Product: skis, skateboard, bike

Recommendation: golf clubs

###

Product: shoes, shirt, hat

Recommendation: Pants

Prompt Engineering Guide

1. Give clearer instructions



2. Split complex tasks into simpler subtasks



3. Structure the instruction to keep the model on task



4. Prompt the model to explain before answering



5. Ask for justifications of many possible answers, and then synthesize



6. Generate many outputs, and then use the model to pick the best one



7. Fine-tune custom models to maximize performance



Prompt Chaining

Entity Extraction

Please **extract entities** from the following news article: 'The new iPhone model is set to be released next month. It has been highly anticipated by Apple fans and is expected to feature a larger screen and improved camera

```
{'text': 'iPhone', 'type': 'PRODUCT'}, {'text': 'Apple', 'type': 'ORGANIZATION'}
```

Summarization

Please **summarize** the information about the product

```
[{'text': 'iPhone', 'type': 'PRODUCT'}, {'text': 'Apple', 'type': 'ORGANIZATION'}]
```

The iPhone is a line of smartphones designed and marketed by Apple Inc. These devices combine a computer, camera, and internet connectivity in a single device, and are known for their sleek design and user-friendly interface.

Sentiment Analysis

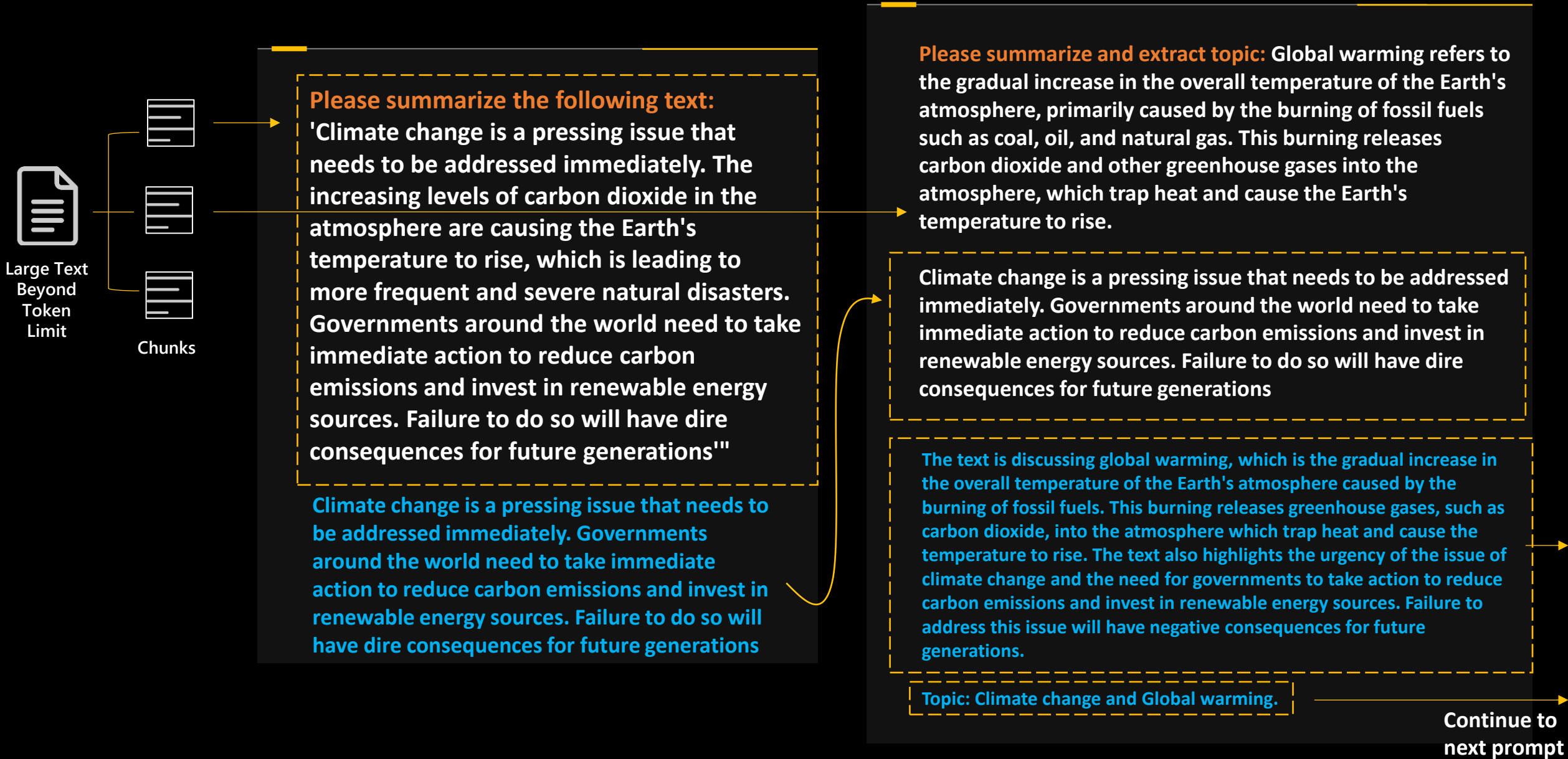
Please provide a **sentiment** for the following text:

The iPhone is a line of smartphones designed and marketed by Apple Inc. These devices combine a computer, camera, and internet connectivity in a single device, and are known for their sleek design and user-friendly interface.

Positive

Prompt Chunking (token limit)

Then all summaries are passed to GPT again to summarize, i.e. second round of summarization. The sum of the summaries has to fit within the token limit.



Few-Shot Reasoning (Human version)

1st prompt

Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: The answer is 11.

The cafeteria has 23 apples. If they used 20 to make lunch and bought 6 more, how many do they have?

The answer is 27



2nd prompt—provide reasoning

Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Answer: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5+6 = 11$. The answer is 11.

The cafeteria has 23 apples. If they used 20 to make lunch and bought 6 more, how many do they have?

The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23-20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9.



Model Reasoning—Chain-of-Thought

Before

What is the annual water demand of a single-family household containing four people who are at home an average of 200 days per year and use an average of 100 liters of water per day?

The annual water demand of a single-family household containing four people who are at home an average of 200 days per year and use an average of 100 liters of water per day is 8,000,000 liters per year.

After

What is the annual water demand of a single-family household containing four people who are at home an average of 200 days per year and use an average of 100 liters of water per day?

Let's think step by step and explain the calculation step by step.

Step 1: Calculate the total number of days in a year:

There are 365 days in a year.

→ **Step 2:** Calculate the total number of days the family is at home:

The family is at home an average of 200 days per year.

→ **Step 3:** Calculate the total water demand of the family:

The family uses an average of 100 liters of water per day. Therefore, the total water demand of the family is $100 \text{ liters} \times 200 \text{ days} = 20,000 \text{ liters per year}$.

→ **Step 4:** Calculate the annual water demand of the single-family household:

The annual water demand of the single-family household is 20,000 liters per year.

Using Chain-of-Thought for Reasoning

1st prompt—Reason Extraction

On average Joe throws 25 punches per minute.
A fight lasts 5 rounds of 3 minutes. How many punches did he throw?

Let's think step by step and explain the calculation step by step.

In one minute, Joe throws 25 punches.
In three minutes, Joe throws $3 * 25 = 75$ punches.
In five rounds, Joe throws $5 * 75 = 375$ punches.

2nd prompt—Answer Extraction

One average Joe throws 25 punches per minute.
A fight lasts 5 rounds of 3 minutes. How many punches did he throw?

In one minute, Joe throws 25 punches.
In three minutes, Joe throws $3 * 25 = 75$ punches.
In five rounds, Joe throws $5 * 75 = 375$ punches.

Therefore, the answer (Arabic numerals) is

The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9.

Few-Shot—Chain-of-Thought—Examples

Math Word Problems (free response)

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$.

The answer is 11.

StrategyQA

Q: Yes or no: Would a pear sink in water?

A: The density of a pear is about 0.6 g/cm^3 , which is less than water. Thus, a pear would float

So the answer is no.

SayCan (Instructing a robot)

Human: How would you bring me something that isn't a fruit?

Explanation: the user wants something to eat that isn't a fruit. An energy bar is not a fruit, so I will bring the user an energy bar.

Plan: 1. find(energy bar) 2. pick(energy bar) 3. find(user) 4. put(energy bar) 5. done().

Math Word Problems (multiple choice)

Q: How many keystrokes are needed to type the numbers from 1 to 500? Answer Choices: (a) 1156 (b) 1392 (c) 1480 (d) 1562 (e) 1788

A: There are 9 one-digit numbers from 1 to 9. There are 90 two-digit numbers from 10 to 99. There are 401 three-digit numbers from 100 to 500. $9 + 90(2) + 401(3) = 1392$.

The answer is (b).

Date Understanding

Q: The concert was scheduled to be on 06/01/1943, but was delayed by one day to today. What is the date 10 days ago in MM/DD/YYYY?

A: One day after 06/01/1943 is 06/02/1943, so today is 06/02/1943. 10 days before today is 05/23/1943.

So the answer is 05/23/1943.

Last Letter Concatenation

Q: Take the last letters of the words in "Lady Gaga" and concatenate them.

A: The last letter of "Lady" is "y". The last letter of "Gaga" is "a". Concatenating them is "ya".

So the answer is ya

CSQA (commonsense)

Q: Sammy wanted to go to where the people were. Where might he go? Options: (a) race track (b) populated areas (c) desert (d) apartment (e) roadblock

A: The answer must be a place with a lot of people. Race tracks, desert, apartments, and roadblocks don't have a lot of people, but populated areas do.

So the answer is (b).

Sports Understanding

Q: Is the following sentence plausible? "Joao Moutinho caught the screen pass in the NFC championship."

A: Joao Moutinho is a soccer player. The NFC championship is part of American football, not soccer.

So the answer is no.

Coin Flip (state tracking)

Q: A coin is heads up. Maybelle flips the coin. Shalonda does not flip the coin. Is the coin still heads up?

A: The coin was flipped by Maybelle. So the coin was flipped 1 time, which is an odd number. The coin started heads up, so after an odd number of flips, it will be tails up.

So the answer is no.

Selection-inference prompting

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slides close in 15 minutes. How many times can she slide before it closes?

14 times



It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slides close in 15 minutes. How many times can she slide before it closes?

To solve "How many times can she slide before it closes" we need to first solve: **Chain-of-Thought**

How much time does it take for Amy to climb and slide?

Decomposes problem into a smaller task (Reasoning)

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slides close in 15 minutes. How many times can she slide before it closes?

How much time does it take for Amy to climb and slide?

It takes Amy 5 minutes to climb and slide

It takes Amy 4 minutes to climb to the top of a slide. It takes her 1 minute to slide down. The water slides close in 15 minutes. How many times can she slide before it closes?

It takes Amy 5 minutes to climb and slide.

She can slide 3 times before the water slides close.

Add new fact to context



Links

Attention is all you need (Transformers)—

[\[1706.03762\] Attention Is All You Need \(arxiv.org\)](#)

Proximal Policy Optimization (PPO)—

[\[1707.06347\] Proximal Policy Optimization Algorithms \(arxiv.org\)](#)